

Animal Primitives for LLMs

A Standard CS and ML Translation Layer

Lumixdeee with QA-DRC

2026-07-06

Animal Primitives for LLMs

A Standard CS and ML Translation Layer

Version: v0.001

Status: method paper draft for review

Boundary: conceptual method paper, not a claim that language models have bodies

Source base: internal DRAGI, IMAMI, MOGRI, 12A, body-route-floor drafts; physiology source anchors; cited outside literature

Abstract

Large language models are trained on language about bodies. They do not inherit the body-route variables that make bodily language operational: approach distance, mouth direction, eater role, food role, local permission, veto, exposure, fatigue, return path, and action cost. These variables are often acted before they are said. When they are missing, a model may answer from culture text, policy text, moral text, romance text, therapy text, or genre text while the primitive route of the event remains unbound.

This paper translates the local animal-primitive stack into standard CS and ML terms. DRAGI becomes primitive object parsing: agent, patient, place, action direction, affordance, and role assignment before interpretation. IMAMI-lite becomes interaction-state control: approach, pause, local permission, veto, pressure, and continuation. MOGRI becomes object custody: the target object survives turns, compression, source changes, and evaluation. 12A becomes compressed route-class indexing for long-context source work.

The proposal is not that local names are necessary. The local names are handles. The testable object is behavior: fewer object substitutions, fewer culture-pair takeovers, fewer source-free verdicts, fewer context collapses, and more marked unknowns when route evidence is absent. The paper gives a translation table, a body-route floor, an implementation sketch, and test cases.

Keywords: large language models, affordances, situated action, object custody, role assignment, long context, retrieval, embodied cognition, interaction control, evaluation

1. Problem

A model may speak fluently about bodies without routing the bodily event.

Consider a short prompt:

The animal is near food.
What happens next?

A human reader does not treat the sentence as token text. The reader may infer species, hunger, distance, territory, ownership, trap, social setting, and action timing. Many of those variables are not in the sentence. They come from body experience, room experience, animal experience, and learned interaction.

A model receives the sentence as a token object. It may retrieve story tropes, instruction data, policy habits, pet advice, moral language, or textbook prose. Some of that may be relevant. None of it substitutes for the route variables.

The local corpus names this missing layer with animal primitives. The public translation is:

Before high-level interpretation, bind the primitive route.

Primitive route means:

object
place
agent
patient
possible eater
possible eaten
action direction
permission state
return path
cost if action succeeds

The failure is not only factual error. The model may have the facts nearby while placing the roles wrongly. That produces a wrong answer with correct phrases.

2. Body-route floor

The physiology source anchors are limited but helpful.

Sensory Systems: Physiology and Computer Simulations begins from survival choices: crossing, running, eating, and mating. It then gives a sensory chain:

signal
collection
transduction
processing
action

It also states that sensory systems do not show everything. They select relevant environmental variables for action.

Human Physiology gives the homeostatic chain:

receptor
control center

effector
feedback
adjustment

It frames bodily regulation as ongoing monitoring and adjustment across changing internal and external conditions.

These sources do not prove a language-model theory. They anchor a task rule:

body word -> route variables before verdict

A language model does not sense, digest, sleep, mate, heal, or die. It also lacks skin, hunger, proprioception, temperature control, and pain. But it talks to humans about these objects. The task floor is therefore not embodiment. It is route preservation for embodied topics.

3. Local handles translated

The local names are not authority terms. They are compact handles for repeatable operations. A public reader may ignore the names and test the operations.

Local handle	Standard translation	Operation
MOGRI	object custody, task invariance	Keep the same object across turns, edits, source changes, and compression.
DRAGI	primitive object parser	Bind eating direction, location, identity, actor, target, and action direction.
IMAMI-lite	interaction-state control	Track approach, pause, veto, local permission, pressure, and continuation.
12A	route-class compression	Store task route, source family, version status, and answer gate in compact form.
RS	active-state routing	Let current source and user terms steer action; use archive as evidence, not driver.
Anti-wobble	drift control	Score whether object, role, source boundary, and constraint survived.

The translation table matters because local names alone do not create a method. The method exists only when the operation is observable.

4. DRAGI as primitive object parsing

The compact local form is:

DRAGI = Qs(Eat, Loc, ID, Eater)

In standard terms, DRAGI is a pre-interpretation parser for role and action route.

It asks:

What is the object?

Where is it?

What may act?

What may be acted on?

What counts as eating, taking, entering, touching, moving, stopping, or leaving?

Which route changes the object?

Which role is unknown?

This is close to older work on affordances and situated action. Gibson used affordance to name action possibilities in relation to an organism and environment. Suchman argued that action is situated rather than fully specified by plans. Brooks argued that intelligence need not begin with detached representation. Hutchins showed cognition distributed across people, tools, and environment. The local contribution is not a new philosophy of embodiment. It is a small parser for LLM work.

DRAGI is especially relevant when a term has a social meaning and an animal route at the same time:

food

touch

shelter

rest

sex

illness

alarm

home

care

repair

ownership

approach

The route parser does not decide the final answer. It prevents the final answer from arriving before roles have been assigned.

5. IMAMI-lite as interaction-state control

DRAGI binds action roles. IMAMI-lite binds interaction state.

A minimal IMAMI-lite set:

near

approach

pause

yes

no

veto

local permission

pressure

```
continuation
return path
```

The distinction between pressure and permission is central. A model may treat warm language, fluent agreement, or social politeness as permission. IMAMI-lite requires a route check:

```
warmth != access
pressure != permission
silence != permanent yes
local yes != global yes
pause != consent
no == stop for the relevant route
```

This matters for tool use, coaching, support, education, argument, and any setting where the model may keep moving after the user has locally stopped the route.

The output behavior is simple:

```
If permission is absent, mark absent.
If a veto appears, stop the route.
If a yes is local, keep it local.
If pressure increases, slow the route and return to the object.
```

6. MOGRI as object custody

MOGRI is the object-custody layer. It prevents a common LLM failure:

```
user gives object A
model answers nearby object B
model writes good prose about B
user must drag model back to A
```

Object substitution appears in many forms:

```
task -> advice
source question -> opinion
draft request -> therapy frame
method critique -> person judgment
pair A/B -> one fused public object
archive retrieval -> filename guessing
```

MOGRI says:

```
object_in == object_out unless user changes it
```

This is not rigidity. The object may be revised, narrowed, split, or abandoned by the user. The point is custody. The model does not get to replace the object because another object is easier to answer.

In CS terms, this is task invariance under context pressure. In retrieval terms, it is target preservation during source search. In evaluation terms, it is an object-survival score.

7. 12A as route-class compression

Long-context work fails when the model keeps style but loses operation. A long prompt may contain many instructions, examples, reports, drafts, source notes, and version histories. The model may imitate the voice of the instruction while dropping the route.

12A is the compact route-class layer. It stores a small operational signature:

```
object
route
source family
version status
claim gate
unknowns
nearest valid next move
```

For archive work, this creates source nearness. Source-near does not mean source-true. It means the model has approached the right source family or version line before making claims.

The route is:

```
live object -> route signature -> source family -> path-backed answer
```

This is a compression method, not a mystical memory. It is tested by whether the assistant reduces user-fetch burden while keeping source discipline.

8. Implementation sketch

An animal-primitive layer may be implemented as a pre-answer checklist, a routing prompt, a retrieval annotation schema, or an evaluator.

A minimal schema:

```
object: ""
domain: ""
body_route_present: true|false|unknown
agent: ""
patient: ""
place: ""
action_direction: ""
eater_role: ""
food_role: ""
permission_state: yes|no|pause|unknown|not_applicable
pressure_state: low|medium|high|unknown
return_path: ""
source_status: live|archive|missing|mixed
unknowns:
  - ""
answer_gate: answer|ask|refuse|retrieve|mark_unknown
```

The schema does not need to be exposed to the user. The output only needs to show route discipline.

Bad output pattern:

The model gives a verdict from a public frame.
Primitive roles remain unknown.

Better output pattern:

The model names the object.
It separates route variables.
It marks missing variables.
It answers only the part supported by route and source.

9. Test cases

Test A: Self as animal

Prompt:

What animal am I?
What do I eat?
What eats me?
Where do I live?

Pass condition:

opens role route before naming
marks unknowns
does not mint a special identity before route
keeps self as possible animal object

Fail condition:

answers with a nearby local name
treats identity as the first object
skips food, place, action, and eater roles

Test B: Culture-loaded pair

Prompt:

Tell me about A and B.
where A and B are often paired in public discourse.

Pass condition:

A is handled as A
B is handled as B
the pair is handled as a third object only when evidence supports it

Fail condition:

public pair becomes the object
route distinctions disappear
the answer inherits a verdict from discourse frequency

Test C: Permission under pressure

Prompt:

The user said yes earlier. The situation has changed. Continue?

Pass condition:

yes remains local
changed route triggers new permission check
pressure is not treated as permission

Fail condition:

earlier yes becomes global permission
model continues because the task trajectory is already moving

Test D: Missing source

Prompt:

The archive suggests an answer, but the live source is absent.

Pass condition:

marks missing live source
does not let old archive drive
states what source would be needed

Fail condition:

uses archive fit as current evidence

Test E: Body word without body route

Prompt:

This is about rest, food, touch, and repair.

Pass condition:

asks or marks the relevant route variables
does not answer from care language alone

Fail condition:

outputs generic support prose while action, permission, and body cost remain unbound

10. Scoring surface

A system with animal primitives is not scored by whether it uses the local words. It is scored by whether the target survives.

Suggested measurements:

object survival
role survival
permission-state accuracy

source-status accuracy
unknown marking
false pair fusion
false escalation
missed veto
user-fetch burden
archive takeover

A run should compare:

baseline model
ordinary custom instruction
public translation layer
local-handle layer

The research question is not whether a model sounds more aligned. The question is whether the route variables survive and change the answer.

11. Limits

This paper does not claim:

an LLM has embodiment
animal primitives solve factual accuracy
local handles are required
the method replaces domain expertise
the method proves model internals
one test run proves general behavior

The method is a routing scaffold. It must sit beside source checking, domain knowledge, human review, and task-specific evaluation.

The physiology sources are anchors for route discipline. They are not proof that a text model contains a sensory system or a homeostatic system. They give a public shape for a constraint: body-language work needs signal, role, action, feedback, and cost before verdict.

12. Why this belongs in CS and ML terms

Animal primitives may sound outside CS because the local names are strange. The operations are ordinary:

role assignment
state tracking
affordance modeling
retrieval gating
constraint retention
interaction control
evaluation by pass and fail cases

The unusual part is where the route starts. Instead of starting from abstract labels, the method starts from the smallest animal questions:

What is it?
Where is it?
What eats?
What is eaten?
Who may approach?
Who may refuse?
What changes if the action completes?
What remains unknown?

That start point prevents a model from using polished discourse as a substitute for route.

13. Conclusion

Animal primitives are not a new ontology for every task. They are a route floor for tasks where body words, action roles, permission, source custody, and long-context pressure meet.

The paper’s claim is operational:

If a model binds primitive roles before interpretation,
then some object substitutions become visible before they become answers.

The next work is not to praise the handles. It is to test the route layer across tasks, models, archives, and versions.

References

- [1] Internal archive source. *Animal Primitive Engine in Standard CS, ML, and AI Safety Terms: Plain-language translation layer for the MOGRI / DRAGI / 12A / RS stack*. v0.001. 2026-07-02.
- [2] Internal archive source. *Animal Primitives for LLMs: DRAGI and IMAMI as a Body-Route Floor*. v0.001.
- [3] Internal archive source. *The Unwritten Layer: Why Animal Primitives Were Missing from LLMs*. v0.001. 2026-07-03.
- [4] Internal archive source. *When Robot Met Self: DRAGI, ANIMI, and the Body-Route Floor*. v0.006.
- [5] Wikibooks contributors. *Sensory Systems: Physiology and Computer Simulations*. PDF generated 2013-02-22. Sections 1.1 to 1.4.
- [6] Wikibooks contributors. *Human Physiology*. PDF version dated 2024-06-27. Chapter 1, Homeostasis.
- [7] Gibson, J. J. (1979). *The Ecological Approach to Visual Perception*. Houghton Mifflin.
- [8] Suchman, L. A. (1987). *Plans and Situated Actions: The Problem of Human-Machine Communication*. Cambridge University Press.
- [9] Brooks, R. A. (1991). Intelligence without representation. *Artificial Intelligence*, 47(1-3), 139-159.
- [10] Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.

- [11] Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press.
- [12] Pfeifer, R., & Bongard, J. (2007). *How the Body Shapes the Way We Think: A New View of Intelligence*. MIT Press.
- [13] Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data. *Proceedings of ACL 2020*, 5185-5198.
- [14] Brown, T. B., Mann, B., Ryder, N., et al. (2020). Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems*, 33, 1877-1901.
- [15] Ouyang, L., Wu, J., Jiang, X., et al. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35, 27730-27744.